ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** 15 m	3/4 = h/20.
3	**(b)** about 400 times farther	Equal angular sizes.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(b)** decreased	Longer shadow means a lower Sun.
6	**(d)** about 3 years	11 days x 3 = 33 days.
7	**(c)** A is true but R is false.	A is true; R is false because it rotates once per revolution.
8	**(b)** high in the sky	Lunar eclipse occurs on a full moon, which is high in the sky at midnight.
9	**(d)** one year	33 x 11 = 363 days.
10	**(c)** A is true but R is false.	A is true; R is false: it reflects sunlight.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	365.25 is slightly more than the actual 365.2422 (1); 1900 is not a leap year but 2000 is (1).	1+1
12	Its lit half faces away from the Earth (1); Dark side faces us (1).	1+1
13	Night side faces different directions (1); Different constellations are visible (1).	1+1
14	Ratio 1.5/2 = h/40 (1); h = 30 m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	At noon: almost no shadow (1); At 3 pm: a shadow pointing east (1); Length is larger than at noon (1).	1+1+1
16	Earth rotates 360 degrees in 24 hours (1); 15 degrees per hour (1); Local noon differs with longitude (1).	1+1+1
17	Circumference about 940 million km (1); Time = 365.25 x 24 x 3600 s = 3.16 x 10^7 s (1); Speed about 30 km/s (1).	1+1+1
	OR: Difference 0.0078 x 100 (1); = 0.78 day (1); Three leap days omitted every 400 years (1).	1+1+1
18	Difference 7.5 degrees (1); 7.5 / 15 = 0.5 hour (1); 30 minutes behind IST (1).	1+1+1
	OR: Equator: Sun high in the sky, short shadow (1); High latitude: Sun lower, longer shadow (1); Tilt and latitude decide (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Measure the shadow of a vertical stick at local noon each week (1); Longest in December (1); Shortest in June (1); Medium at equinoxes (1); Graph shows a smooth annual cycle (1).	1+1+1+1+1
	OR: Latitude = 30 degrees N (1); Equator: on the horizon (1); North Pole: overhead (1); Height increases as he travels north (1); Pole Star altitude equals latitude (1).	1+1+1+1+1
20	Divisible by 4 is leap (1); Century years need divisibility by 400 (1); 1800, 1900, 2100 not leap (1); 2000 and 2024 leap (1); Average year length 365.2425 days (1).	1+1+1+1+1
	OR: Orion: winter, belt of three stars (1); Ursa Major: seven stars in a dipper (1); Cassiopeia: W shape near the Pole Star (1); Seasonal visibility changes due to revolution (1); Pole Star near them is fixed (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 15 days	Half month.
2	**(b)** the northern hemisphere is tilted towards the Sun	Tilt towards the Sun.
3	**(c)** A is true but R is false.	A is true; R is false.
4	**(d)** 7 days	A quarter of a month more.
5	**(b)** lowest in the sky	Shortest day.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** 52 weeks and 1 day	365 = 52 x 7 + 1.
8	**(c)** June	Southwest monsoon.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(c)** Monday	365 days gives 1 extra day.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sun is about 400 times larger (1); Also about 400 times farther away (1).	1+1
12	Shadow length changes slowly near noon (1); Hard to mark the shortest shadow (1).	1+1
13	It is an optical illusion (1); Size is the same (1).	1+1
14	Actual month is 29.53 days (1); Months alternate to average out (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	100 = 14 x 7 + 2 (1); Add 2 days to Monday (1); Wednesday (1).	1+1+1
16	Total: Moon fully covers the Sun (1); Annular: Moon is farther away and appears smaller (1); A ring of the Sun remains visible (1).	1+1+1
17	Moon is very far away (1); Angle changes very little as we move (1); Hence it appears to keep pace (1).	1+1+1
	OR: Measure the altitude of the Pole Star (1); Latitude equals the altitude (1); Use a clinometer or an astrolabe (1).	1+1+1
18	Moon blocks the Sun (1); Umbra reaches the Earth as a small spot (1); Spot moves along a narrow path (1).	1+1+1
	OR: Observe the Moon's phases (1); Count days from new moon (1); Festivals mark dates (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Phases: viewing angle of the lit half (1); Eclipses need exact alignment (1); Solar eclipse at new moon (1); Lunar eclipse at full moon (1); Moon's orbit is tilted so eclipses are rare (1).	1+1+1+1+1
	OR: Gnomon fixed pointing north-south (1); Mark the shadow positions hourly on a sunny day (1); Needs sunlight to cast a shadow (1); Reads local solar time (1); Cannot show seconds accurately (1).	1+1+1+1+1
20	Indian: lunisolar, adjusted by Adhik maas (1); Diwali stays within Oct-Nov (1); Islamic: purely lunar, 354 days (1); Eid moves about 11 days earlier each year (1); Gregorian: solar, fixed (1).	1+1+1+1+1
	OR: Day would be 12 hours (1); Days in a year would double (about 730) (1); Seasons unchanged (1); Moon's phases unchanged (1); Moon's rotation is slow so lunar day about 29.5 Earth days (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** earlier in the night	Earth's orbital motion.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(c)** summer	Summer solstice region.
4	**(d)** daily rise and set of Sun and stars	Apparent motion.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(b)** 27 degrees	Altitude equals latitude.
7	**(c)** Sun, Earth and Moon are nearly in a line	Alignment.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** overhead	Equinox.
10	**(b)** does not set for a day	Midnight sun.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They are not on the axis line (1); They trace circles around the pole (1).	1+1
12	The Earth rotates during the exposure (1); Stars appear to move in circles (1).	1+1
13	31 + 29 + 31 + 15 (1); = 106 days (1).	1+1
14	Near equinoxes the Sun's path changes quickly (1); Near solstices the change is slow (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fix a viewing point and mark the position of a star (1); Look again after one hour (1); The star has moved along an arc around the Pole Star (1).	1+1+1
16	Earth lies between the Sun and Moon (1); Moon passes into the Earth's shadow (1); Totality up to about an hour or more (1).	1+1+1
17	Helped in observing the Sun and stars (1); Use of the shadow of a gnomon to find directions (1); Alignment gave symbolic and practical use (1).	1+1+1
	OR: Earth moves around the Sun (1); Line of sight to the Sun changes (1); Sun appears to travel through constellations once a year (1).	1+1+1
18	Seven bodies visible to the naked eye (1); Each given a day (1); Used in calendars (1).	1+1+1
	OR: Rotation periods differ (1); Moon rotates slowly (1); Days are longer there (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	At the equator, they rise and set vertically (1); At the poles, they circle horizontally (1); Pole Star altitude equals latitude (1); Axis direction fixed (1); Rotation causes apparent motion (1).	1+1+1+1+1
	OR: Moon blocks the Sun (1); Umbra falls on a narrow path (1); Rare at one location (1); Safety: filters (1); Corona and diamond ring (1).	1+1+1+1+1
20	Aryabhata: rotation of the Earth (1); Varahamihira: Panchasiddhantika (1); Brahmagupta: eclipse calculation (1); Jai Singh: observatories (1); Calendars and almanacs (1).	1+1+1+1+1
	OR: Observe the Moon daily at the same time (1); Draw its shape and record the date (1); Note rise/set times (1); Group into phases (1); Present as a chart and link with tithi (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option

attempted. Deduct up to ½ mark for missing units or figures where required.

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